

Exp no: 7

Roll no: 241001056

Unification and Resolution – Exercises with Solutions

Exercise 1: Sentence Matcher

% Facts

sentence(the_cat_sits).

sentence(the_dog_runs).

% Query

% ?- sentence(X).

Output:

X = the_cat_sits ; X = the_dog_runs ; false.

Exercise 2: Compound Term Unification

% Facts

likes(john, food(pizza, spicy)).

likes(mary, food(pizza, sweet)).

% Query

% ?- likes(john, food(pizza, T)).

Output:

T = spicy.

Exercise 3: Conflict Resolution

% Facts

rule1(climate_change).

rule2(climate_change).

% Rule

conflict(X) :- rule1(X), rule2(X).

% Query

% ?- conflict(climate_change).

Output:

true.

% Query

SWI-Prolog console

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Please run `?- license.` for legal details.

For online help and background, visit <https://www.swi-prolog.org>

For built-in help, use `?- help(Topic).` or `?- apropos(Word).`

12 `?- consult(['C:/Users/Lenovo/Documents/er.pl']).`

true.

13 `?- likes(john, food(pizza, T)).`

`T = spicy.`

14 `?- consult(['C:/Users/Lenovo/Documents/ep.pl']).`

true.

15 `?- sentence(X).`

`X = the_cat_sits .`

16 `?- onconsult(['C:/Users/Lenovo/Documents/eq.pl']).`

Correct to: `"consult(['C:/Users/Lenovo/Documents/eq.pl'])"`?

Please answer 'y' or 'n'? yes

true.

17 `?- conflict(climate_change).`

true.

18 `?-` ♦

Result:

Thus the programs in prolog has been executed and the output is verified successfully.